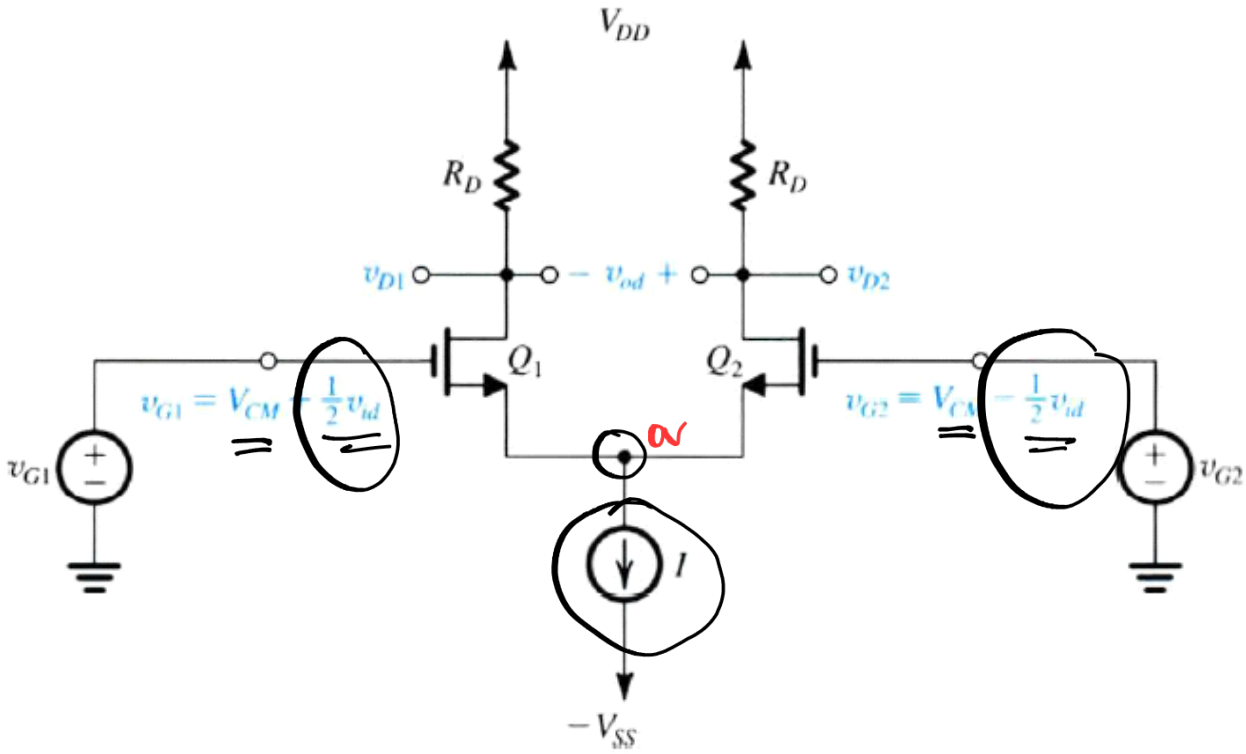
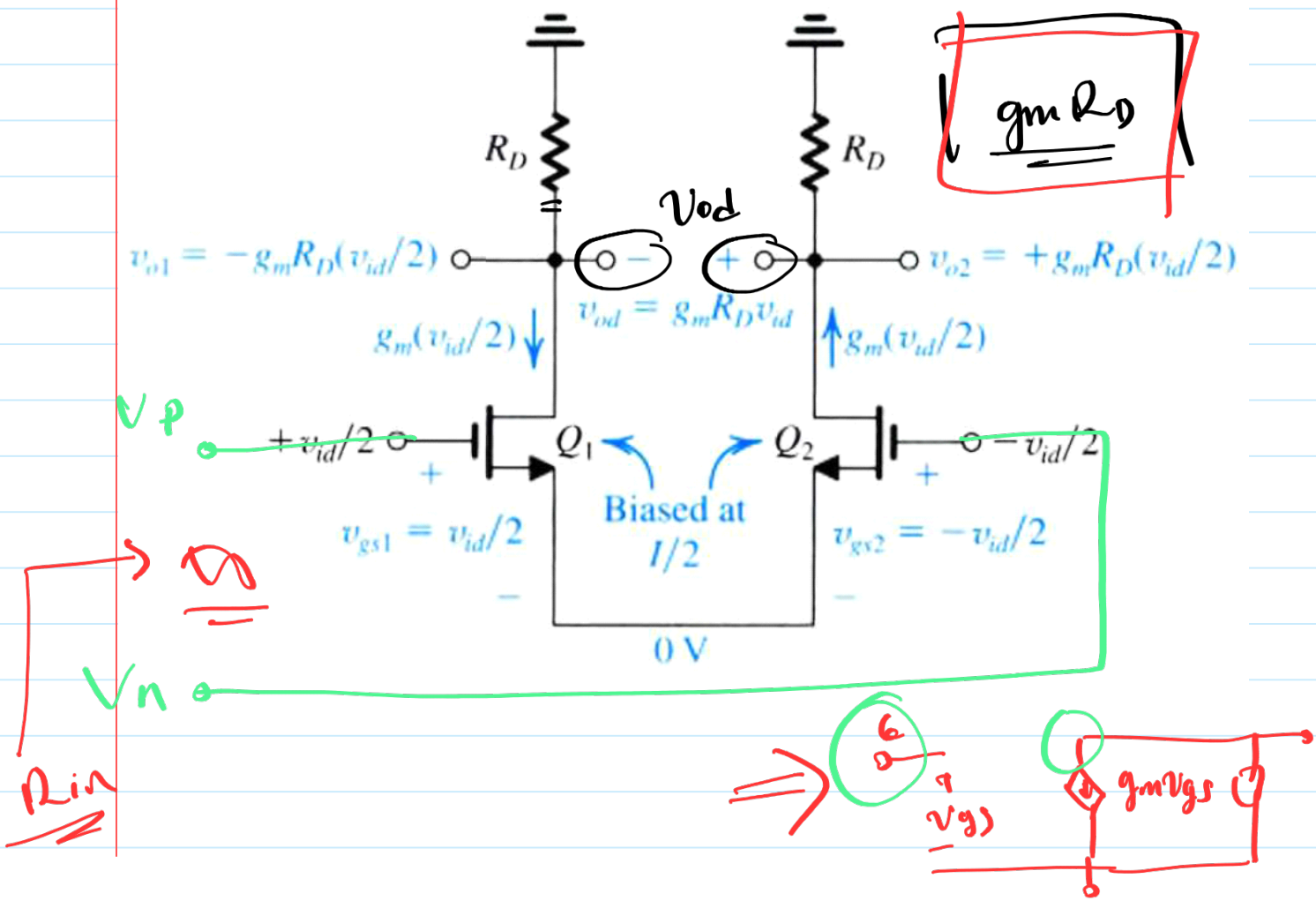
# Reading

Adel S. Sedra and Kenneth C. (KC) Smith, Tony Chan Carusone, and Vincent Gaudet
**Microelectronic Circuits** 8th Edition, *Oxford University Press*, 2019.

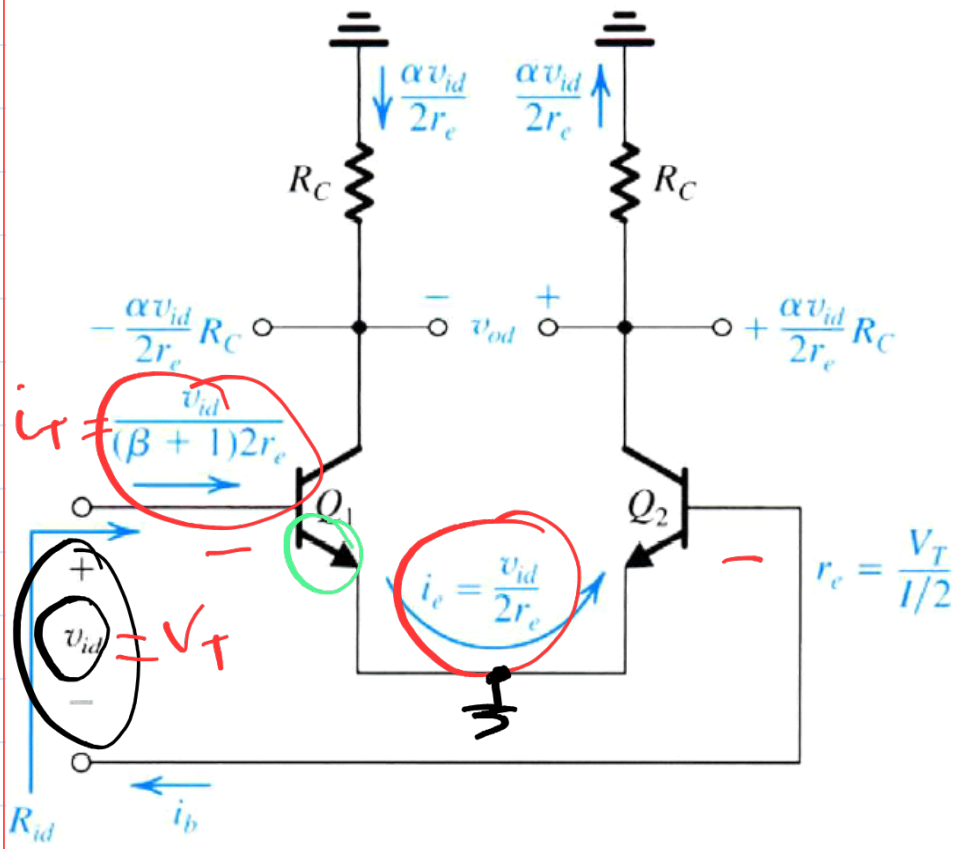
- Chapter 9.3 Common-Mode Rejection
- Chapter 9.4 DC Offset

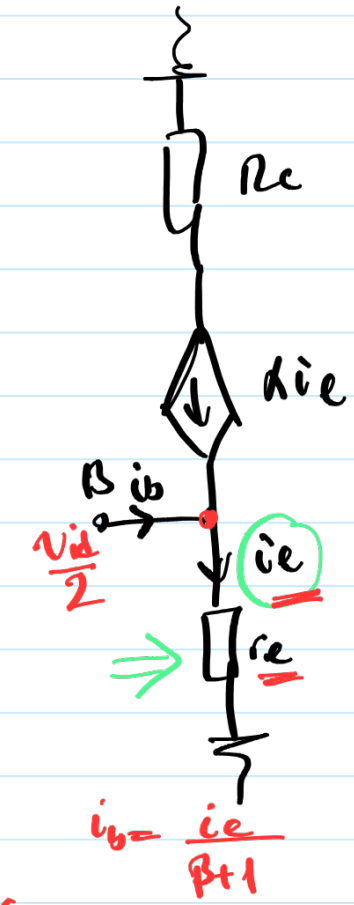
# Recap


$V_{DD}$
$R_D$
$R_D$
$v_{D1}$
$v_{od}$
$v_{D2}$
$Q_1$
$Q_2$
$v_{G1} = V_{CM} + \frac{1}{2}v_{id}$
$v_{G2} = V_{CM} - \frac{1}{2}v_{id}$
$v_{G1}$
$v_{G2}$
$I$
$-V_{SS}$

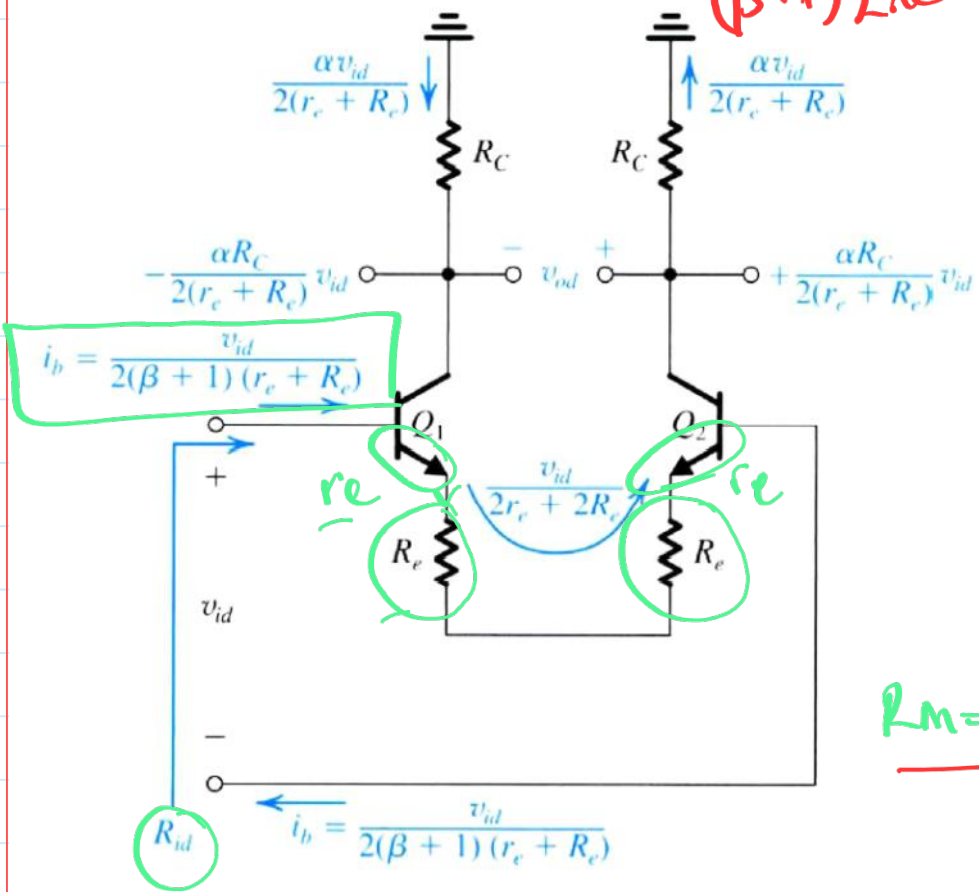

$R_D$
$R_D$
$v_{o1} = -g_m R_D(v_{id}/2)$
$v_{o2} = +g_m R_D(v_{id}/2)$
$v_{od} = g_m R_D v_{id}$
$g_m(v_{id}/2)$
$g_m(v_{id}/2)$
$+v_{id}/2$
$-v_{id}/2$
$Q_1$
$Q_2$
Biased at $I/2$
$v_{gs1} = v_{id}/2$
$v_{gs2} = -v_{id}/2$
0 V

# Input Differential Resistance



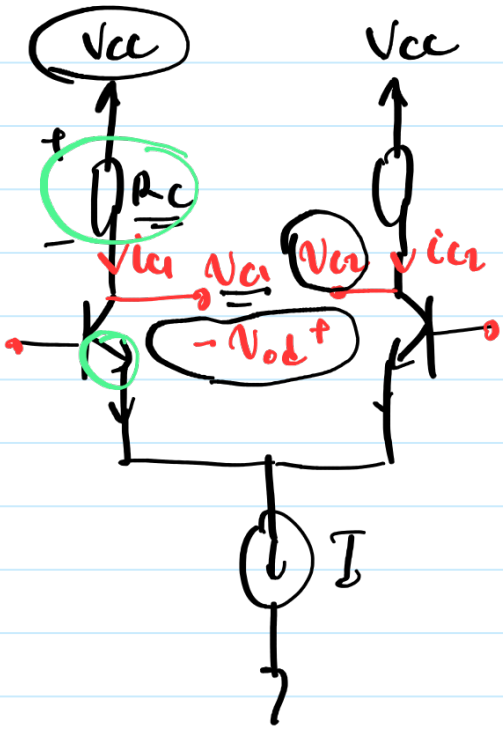

$$R_{in} = \frac{V_T}{i_T} = \frac{v_{id}}{\frac{v_{id}}{(\beta+1)2r_e}} = (\beta+1)2r_e = 2r_\pi$$




$$R_{in} = \frac{v_{id}}{\frac{v_{id}}{(re+Re)2(\beta+1)}}$$

$$R_{in} = (\beta+1)2(re+Re)$$

# Differential Voltage Gain




$$i_{c1} = I_c + g_m \frac{v_{id}}{2}$$

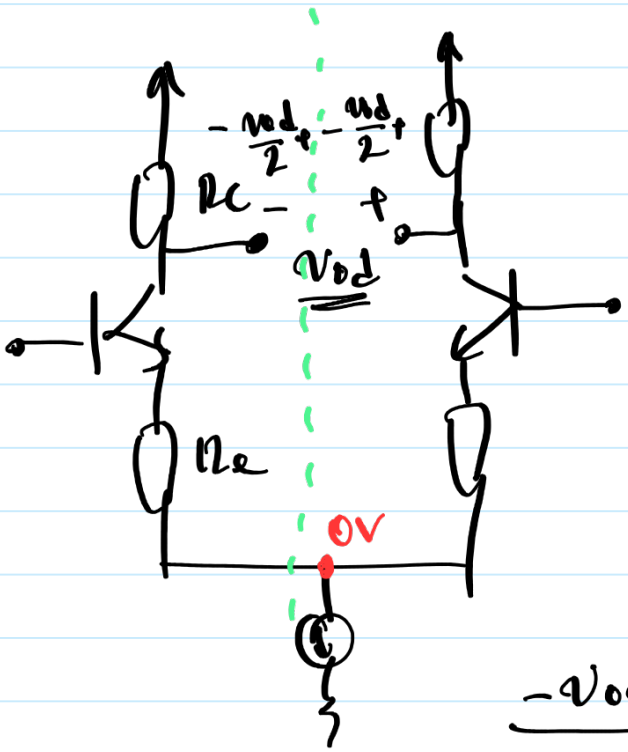
$$i_{c2} = I_c - g_m \frac{v_{id}}{2}$$

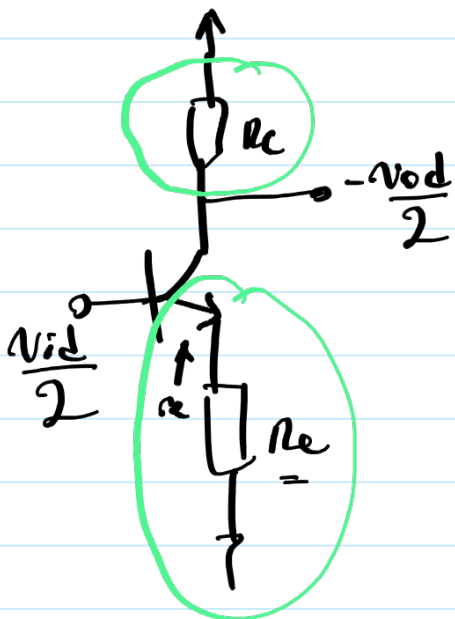
$$v_{c1} = (V_{cc} - I_c R_c) - g_m R_c \frac{v_{id}}{2}$$

$$v_{c2} = (V_{cc} - I_c R_c) + g_m R_c \frac{v_{id}}{2}$$

$$v_{od} = v_{c2} - v_{c1}$$

$$v_{od} = g_m R_c v_{id} \Rightarrow \frac{v_{od}}{v_{id}} = g_m R_c = \frac{R_c}{r_e}$$

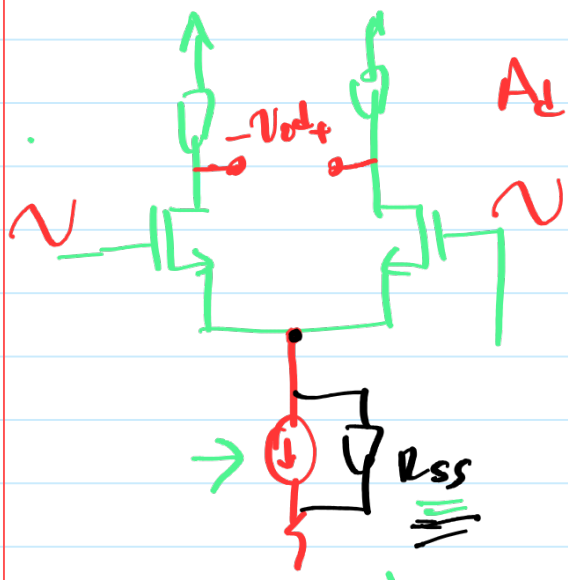


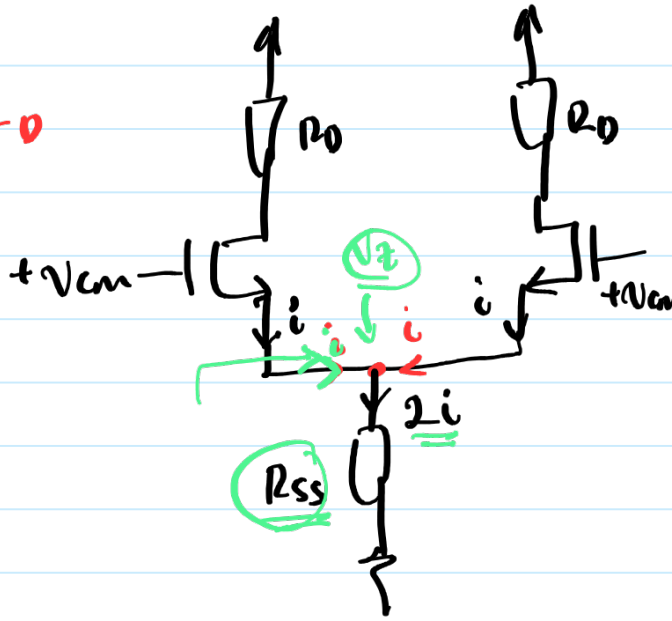

$$\frac{-v_{od}/2}{v_{id}/2} = \frac{R_c}{r_e + R_e}$$

# Differential Half Circuit

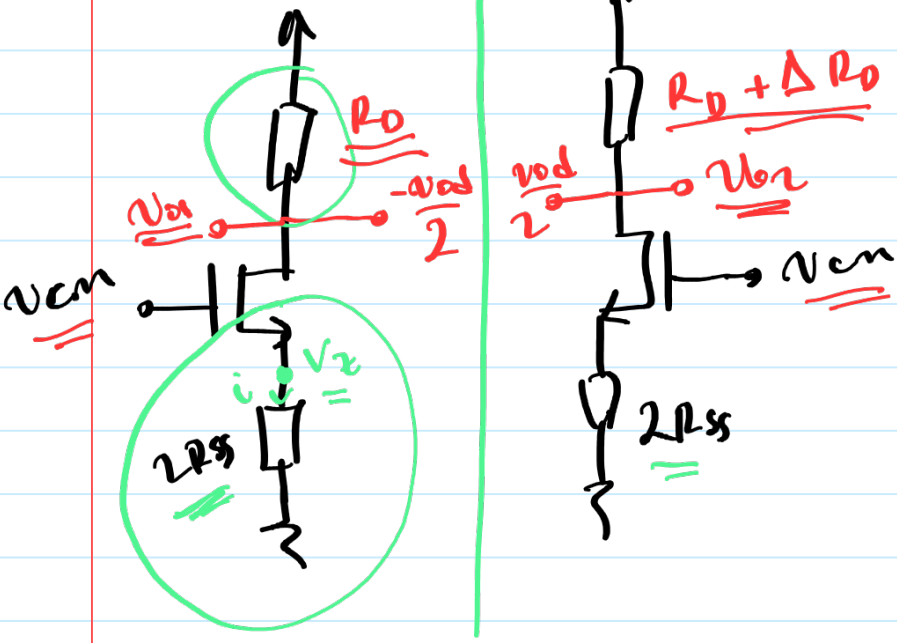
# Common Mode Rejection (MOS)

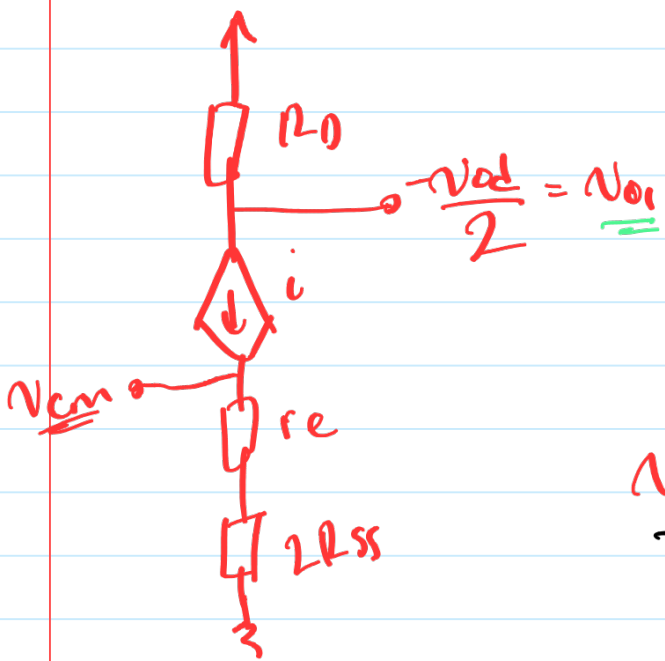



$A_d = g_m R_D$




$\frac{V_t}{2R_{ss}} = i$




$$V_{o1} = \frac{-R_D}{r_e + 2R_{ss}} \cdot V_{cm}$$

$$V_{o2} = \frac{-R_D}{r_e + 2R_{ss}} \cdot V_{cm}$$

$$V_{od} = V_{o2} - V_{o1} = 0$$

# Common Mode Rejection (MOS)

$$V_{o1} = \frac{-R_D}{2R_{SS}} V_{cm}$$

$$V_{o2} = -\frac{(R_D + \Delta R_D)}{2R_{SS}} \cdot V_{cm}$$

$$V_{od} = V_{o2} - V_{o1} = -\frac{\Delta R_D}{2R_{SS}} \cdot V_{cm}$$

$$A_{cm} = \frac{V_{od}}{V_{cm}} = -\frac{\Delta R_D}{2R_{SS}} = -\left(\frac{R_D}{2R_{SS}}\right)\cdot\left(\frac{\Delta R_D}{R_D}\right)$$

$+\frac{V_{id}}{2}$ $\quad -\frac{V_{id}}{2}$ $\rightarrow$ $\frac{V_{od}}{V_{id}} = A_d$

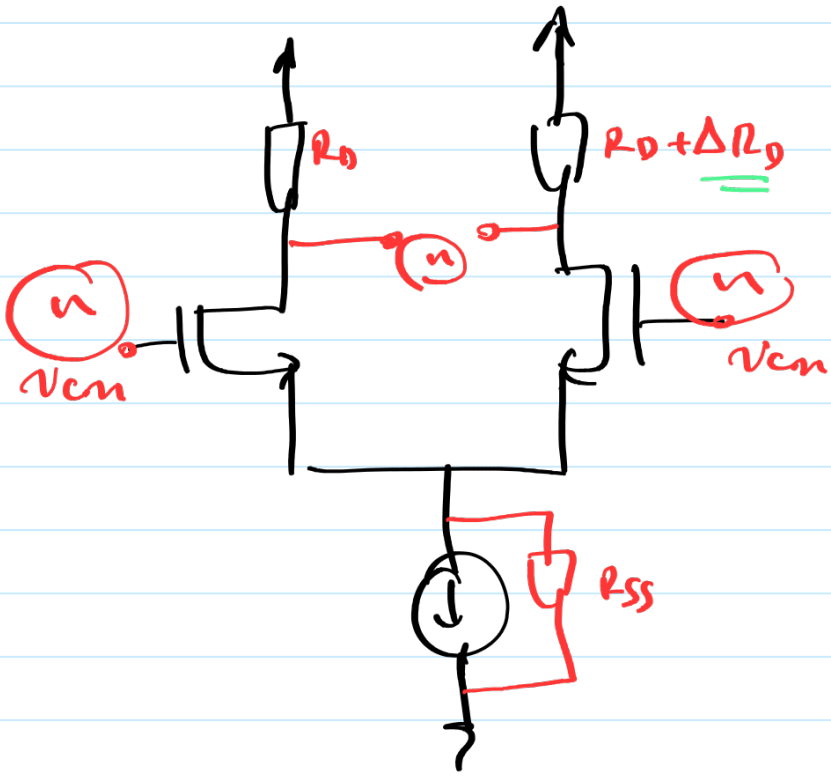
$+V_{cm}$ $\quad +V_{cm}$ $\rightarrow$ $\frac{V_{od}}{V_{cm}} = A_{cm}$

$$CMRR = \frac{|A_d|}{|A_{cm}|}$$

$$CMRR(dB) = 20 \log \frac{|A_d|}{|A_{cm}|}$$

$$CMRR = \frac{g_m R_D}{\Delta R_D / 2R_{SS}} = g_m 2R_{SS} \Big/ \left(\frac{\Delta R_D}{R_D}\right)$$

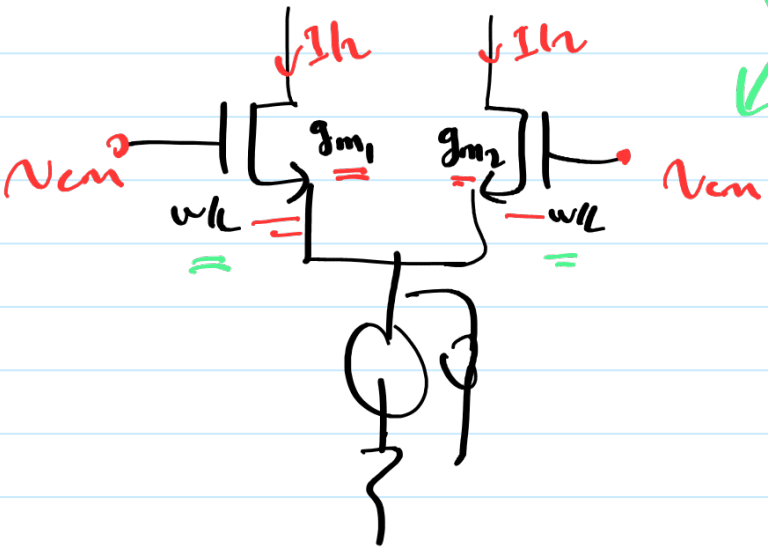
# Effect of RD Mismatch




$$CMRR = \frac{|A_d|}{|A_{cm}|}$$

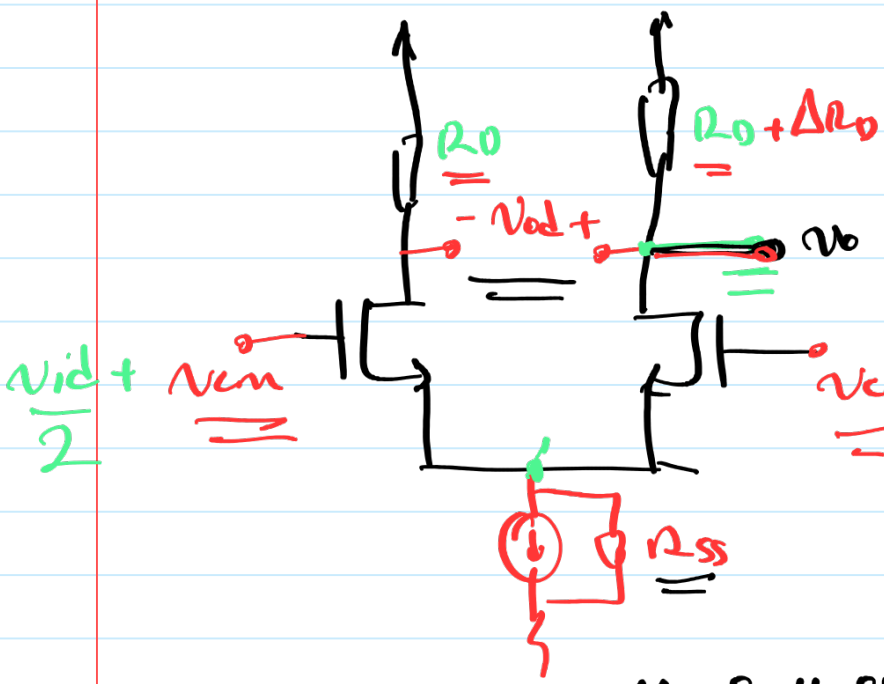
$$A_{cm} = \left(\frac{R_D}{2R_{SS}}\right)\cdot\left(\frac{\Delta R_D}{R_D}\right)$$

# Effect of gm Mismatch




$$A_{cm} = \left(\frac{R_D}{2R_{SS}}\right)\left(\frac{\Delta g_m}{g_m}\right)$$

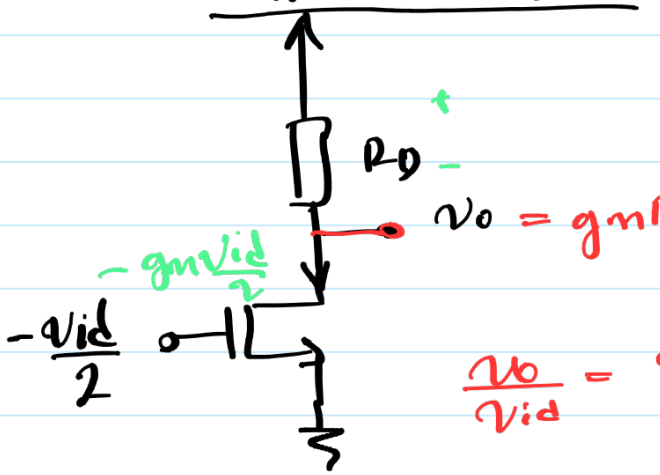
# Single-Ended Output




$$A_d = g_m R_D$$

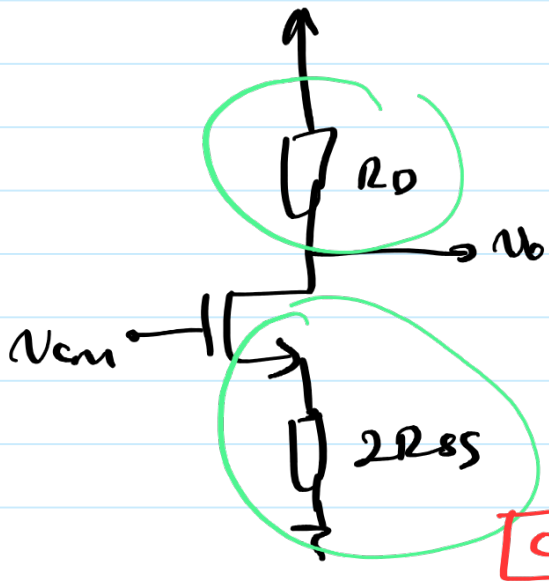
$$A_{cm} = \left(\frac{R_D}{2R_{SS}}\right)\left(\frac{\Delta R_D}{R_D}\right)$$

Diff Small Sig Ckt.




$$\frac{v_o}{v_{id}} = \frac{g_m R_D}{2} = A_d$$

Common Mode Small Sig.



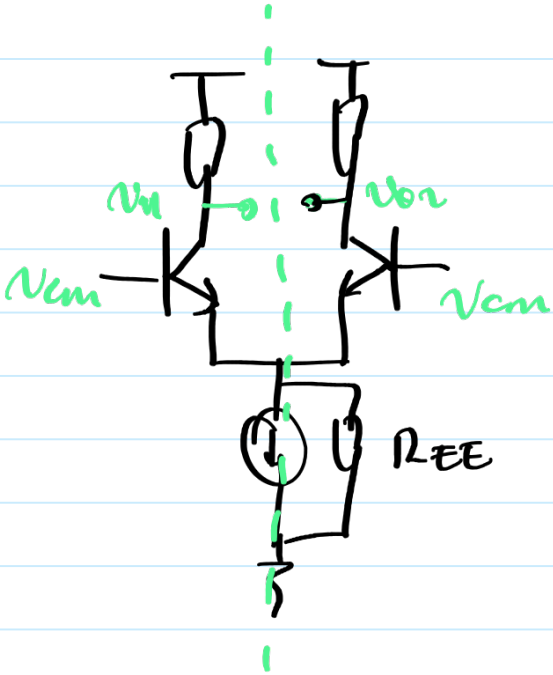

$$A_{cm} = -\frac{R_D}{2R_{SS}}$$

$$CMRR = \frac{|A_d|}{|A_{cm}|}$$

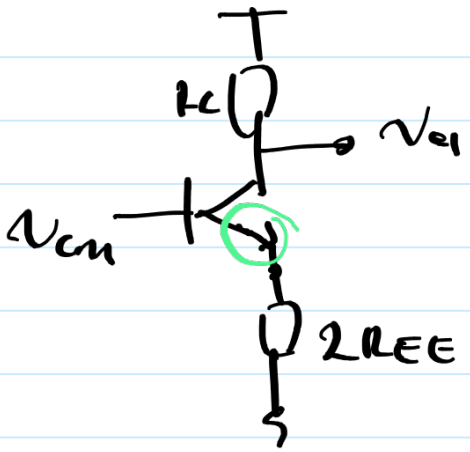
$$= \frac{g_m R_D}{2} \cdot \frac{2R_{SS}}{R_D}$$

$$CMRR = g_m R_{SS}$$

# Common Mode Rejection (BJT)




$$\frac{v_{o1}}{v_{cm}} = ?$$

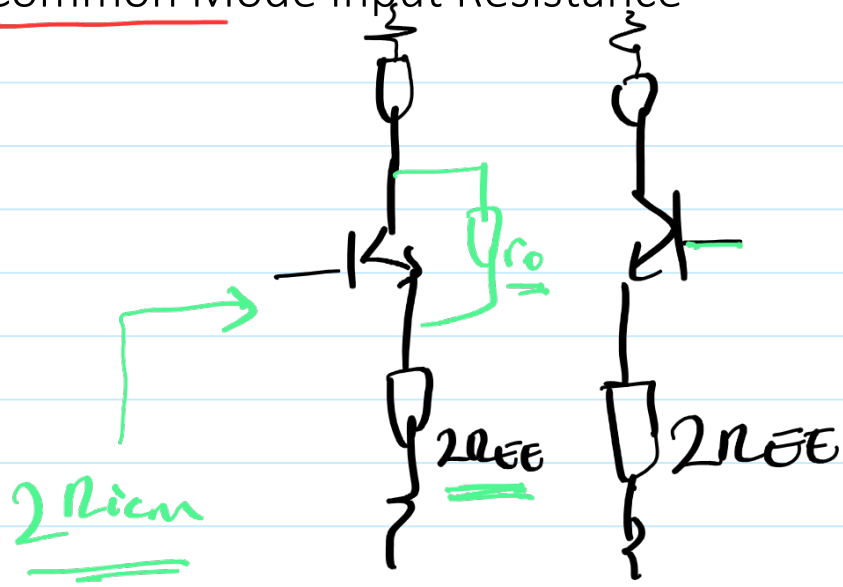



$$v_{o1} = -\frac{\alpha R_C}{r_e + 2R_{EE}}$$

$$v_{o1} = v_{o2} \Rightarrow v_{od} = v_{o2} - v_{o1}$$

$$A_{cm} = -\left(\frac{R_C}{2R_{EE}}\right) \cdot \left(\frac{\Delta R_C}{R_C}\right)$$

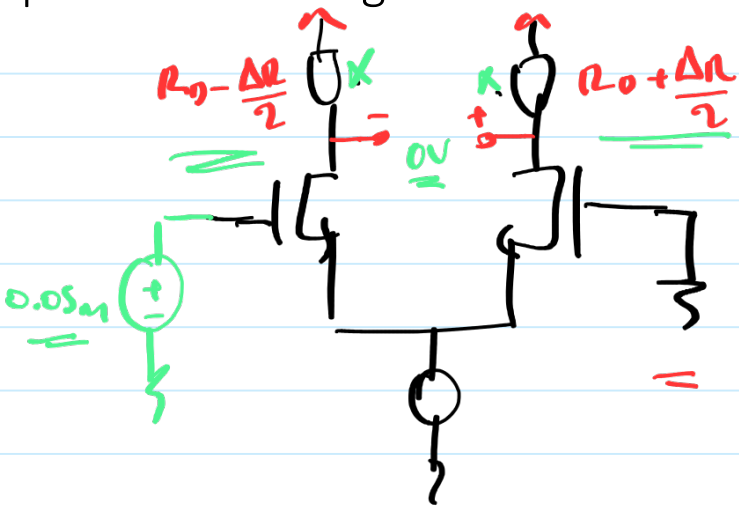
# Common Mode Input Resistance




$$R_{icm} \overset{u}{=} R_{EE}\,(\beta+1)$$

$$R_{icm} \overset{u}{=} \beta R_{EE}\,\frac{(1+R_C/\beta r_o)}{1+\dfrac{R_C+2R_{EE}}{r_o}}$$

# Input Offset Voltage of MOS DiffAmp



# Input Offset Voltage of BJT DiffAmp

Template